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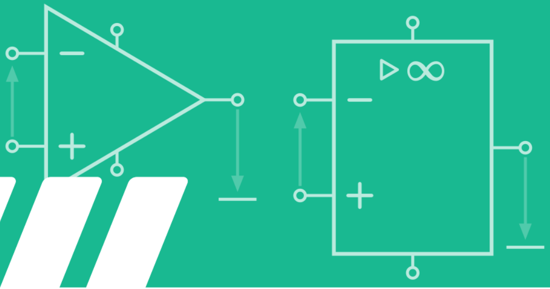
Modul 10: Operational amplifier



Stand: 18. Februar 2026

License

Learning objectives: Operational amplifier

The students can

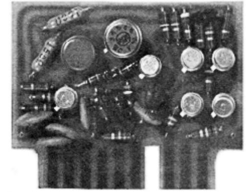
- Describe the structure of a simple operational amplifier.
- Explain the operating principle of an operational amplifier.
- Name different operational amplifier circuits and possible areas of application.
- Describe the differences between the simplified operational amplifier model and real operational amplifiers and specify important areas of the operational amplifier characteristic curve.
- Determine the function of existing operational amplifier circuits and calculate the gain using Kirchhoff's laws.

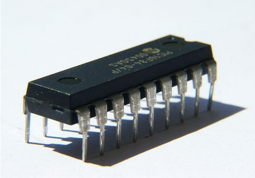
Learning objectives: Operational amplifier

The students can

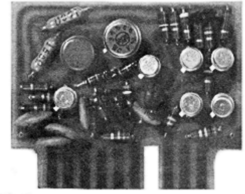
- Specify suitable operational amplifier circuits for a problem solution, calculate resistance ratios, and record the circuit.
- Name important properties of operational amplifier circuits.
- Perform stability analyses and evaluate the results of the analysis.
- Based on data sheets, name the advantages and disadvantages of operational amplifiers for specific applications and select suitable components for the problems at hand.

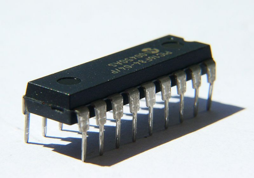
History and structure





- Multi-stage transistor amplifiers can be used to amplify signals (see chapter on semiconductors).





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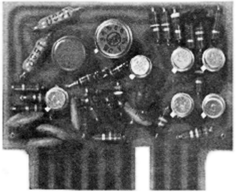
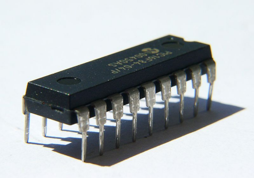
- Circuits designed according to this principle were constructed discretely as universal amplifiers until the development of integrated circuits (ICs).

History and structure





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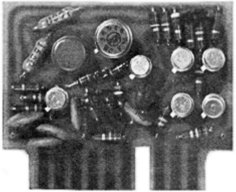
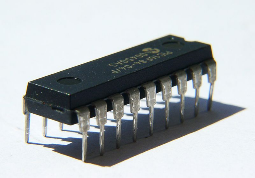
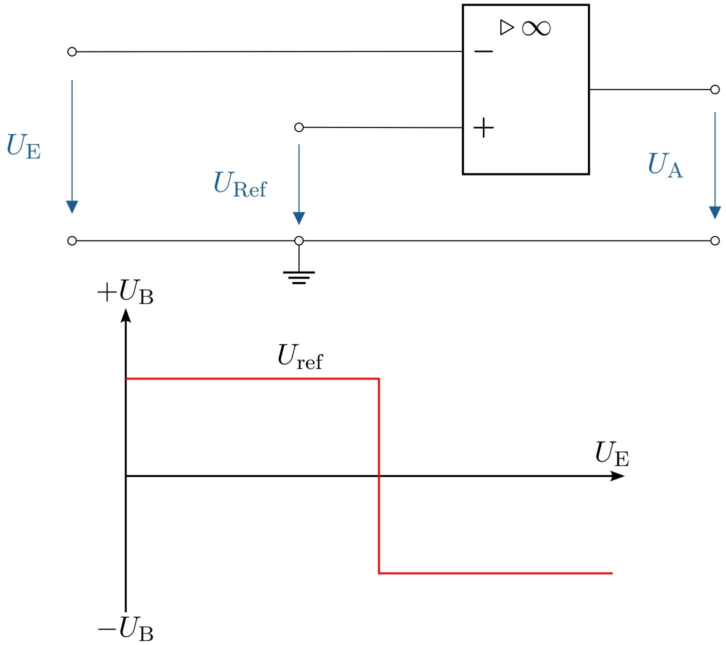
- Today, universal amplifiers are constructed monolithically. Their behavior for the desired application is largely determined by the external circuitry.

History and structure



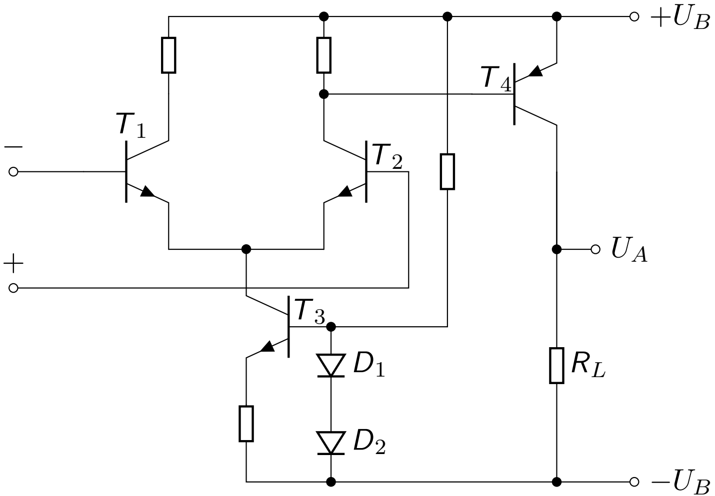


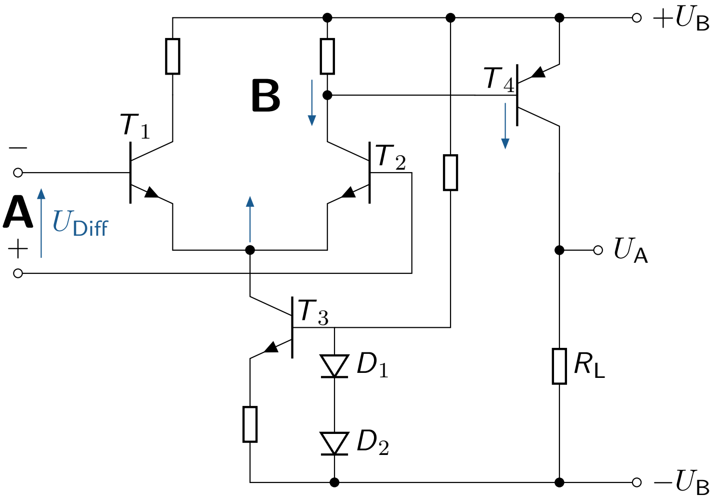
- Multi-stage transistor amplifiers can be used to amplify signals (see chapter on semiconductors).
- Circuits designed according to this principle were constructed discretely as universal amplifiers until the development of integrated circuits (ICs).
- Today, universal amplifiers are constructed monolithically. Their behavior for the desired application is largely determined by the external circuitry.
- They are usually referred to as „operational amplifiers“ due to their original use in analog computers.



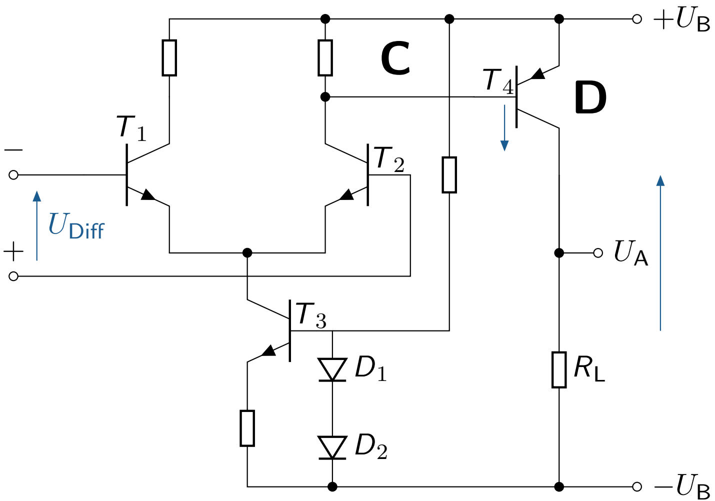
$\triangleright\ \infty$
$-$
$+$
U_E
U_Ref
U_A
$+U_\mathrm{B}$
U_ref
U_E
$-U_\mathrm{B}$

The following circuit illustrates how this works. What happens when a positive differential voltage is applied to this circuit?

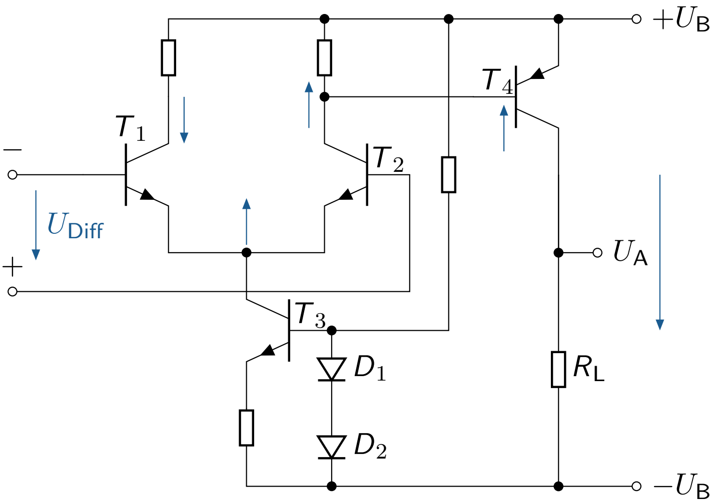




A positive differential voltage (A) leads to a reduction in the resistance of T_2, which results in a voltage reduction at its collector and an increase at the emitter (B). T_3 acts as a current source.

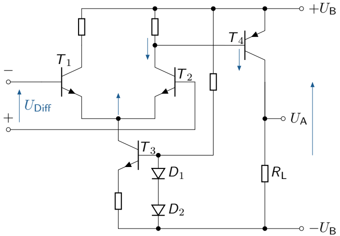


Due to the voltage increase at the base of T_4 (C), the collector-emitter path of T_4 conducts more current (D), which leads to an increased voltage drop across the load resistor R_L and thus to a higher output voltage U_A.

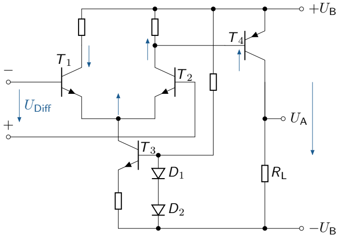


A higher inverted input voltage, on the other hand, reduces the resistance in the collector-emitter path of T_1. The resulting increase in resistance of T_4 reduces the current flow through R_L, which also causes the voltage U_A to drop.

Important characteristics of the ideal amplifier



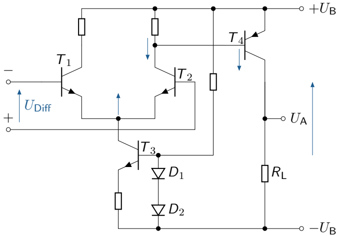
Amplifier with positive differential voltage



Amplifier with negative differential voltage

This circuit already reveals some important characteristics of the real operational amplifier:

- The input current consumption of the amplifier depends on transistors T_1 and T_2.



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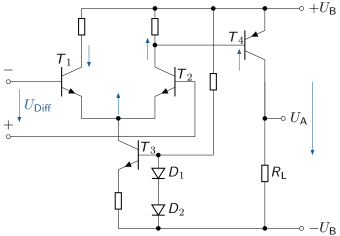

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- The input current consumption of the amplifier depends on transistors T_1 and T_2.
- The amplification is limited by the supply voltage at the top and bottom.

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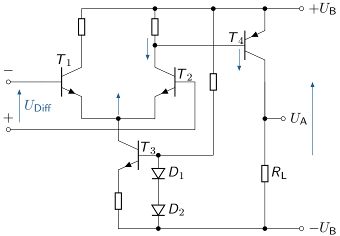

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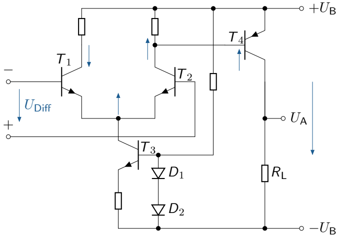
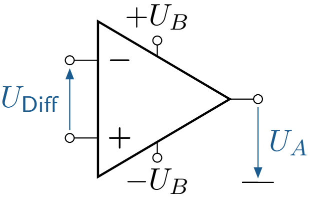


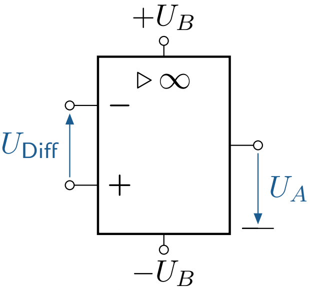
Amplifier with negative differential voltage

This circuit already reveals some important characteristics of the real operational amplifier:

- The input current consumption of the amplifier depends on transistors T_1 and T_2.
- The amplification is limited by the supply voltage at the top and bottom.
- The available output current is determined by the source and transistor T_4.

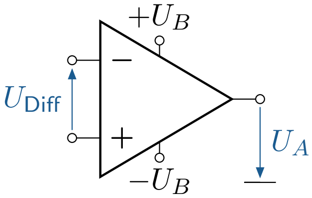


old circuit symbol



new circuit symbol

- In circuits, the operational amplifier is represented by one of the two symbols shown on the left.



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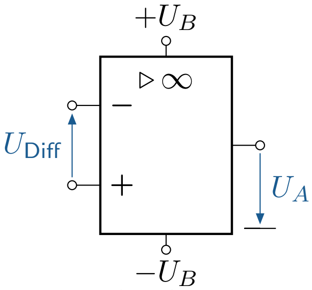

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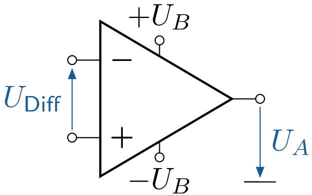
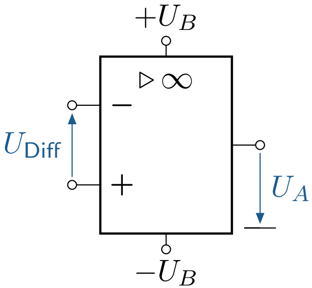
- In circuits, the operational amplifier is represented by one of the two symbols shown on the left.
- The operational amplifier can thus be modelled using a so-called two-port network.



old circuit symbol



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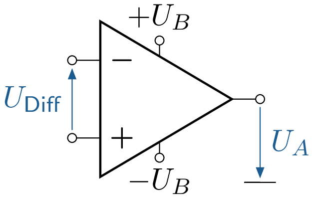
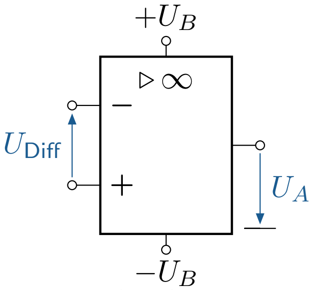
- In circuits, the operational amplifier is represented by one of the two symbols shown on the left.
- The operational amplifier can thus be modelled using a so-called two-port network.
- This is characterised by the fact that, in simplified terms, it can be assumed that the transfer behaviour can be described entirely by the variables current and voltage.



old circuit symbol



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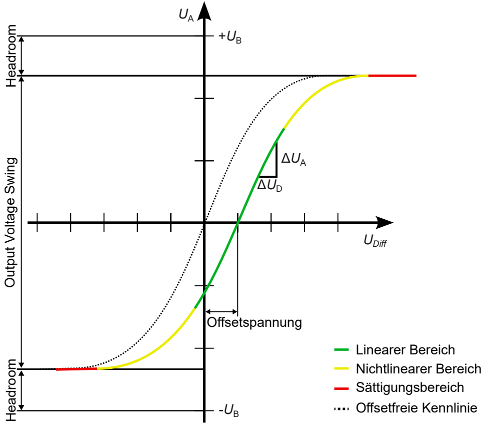
- The operational amplifier can thus be modelled using a so-called two-port network.
- This is characterised by the fact that, in simplified terms, it can be assumed that the transfer behaviour can be described entirely by the variables current and voltage.
- This simplification allows the internal structure of the operational amplifier to be disregarded and treated as a 'black box'.

Characteristic values

Designation	**Ideal OPV-properties**	**Typical values (e.g. OPA 121)**
Open-circuit gain	$V_{\text{Leer}} = \infty$	$V_{\text{Leer}} = 10^6$
Input impedance	$Z_{\text{i}} = \infty\ \Omega$	$Z_i = 10^{13}\ \Omega$
Output impedance	$Z_{\text{a}} = 0\ \Omega$	$Z_{\text{a}} = 50$ bis $100\ \Omega$
Bandwidth	$B = \infty$ MHz	$B > 2$ MHz
Phase shift	$\varphi = 0°$	$\varphi > 0°$
Slew Rate	∞ V	2 mV up to 1000 V/μs
Output controllability	∞	Rail-to-Rail
Input quiescent current	$I_{\text{Ruhe}} = 0$ pA	$I_{\text{Ruhe}} < 5$ pA
Input offset current	$I_{\text{Off}} = 0$ pA	$I_{\text{Off}} < I_+ - I_-$
Input offset voltage	$U_{\text{Off}} = 0$ mV	$U_{\text{Off}} < 2$ mV
Common mode-Rejection	$CMR = \infty$ dB	$CMR = 86$ dB

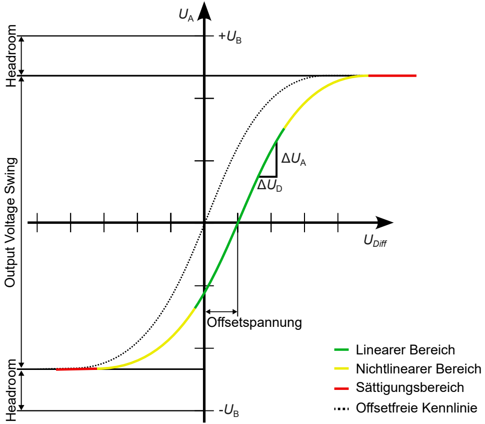
General characteristic curve of an operational amplifier



Characteristic curve of an operational amplifier

- U_{Off} ist die Offset-Spannung, die zu einer horizontalen Verschiebung der Kennlinie führt.

General characteristic curve of an operational amplifier



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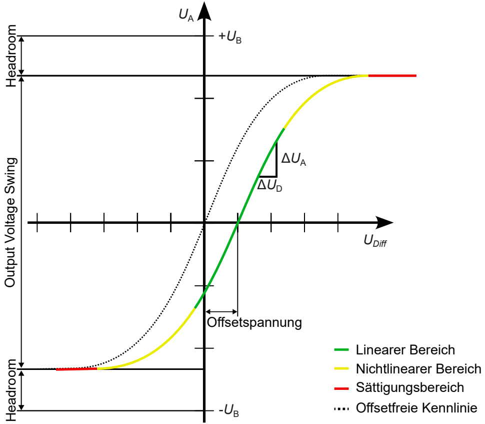

Characteristic curve of an operational amplifier

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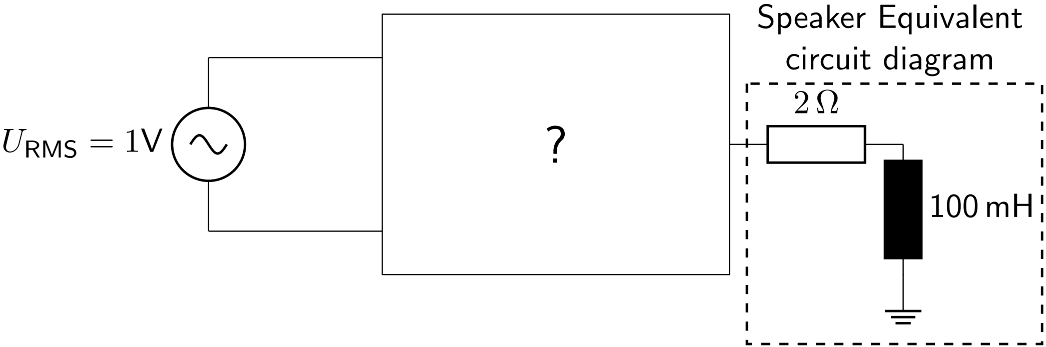
output voltage range.
- The output voltage range is-specified in the data sheet by the value „Output Voltage Swing".

Principle of negative feedback

A power amplifier is to be constructed that is capable of amplifying the signal from a mobile phone so that it can be used to operate a loudspeaker. The input voltage level is 1 V and a 4 Ω loudspeaker is to be operated. How high must the voltage at the output of the amplifier be so that the loudspeaker absorbs 25 W at 1 kHz? How can this be achieved if an operational amplifier circuit is to be used for the solution?



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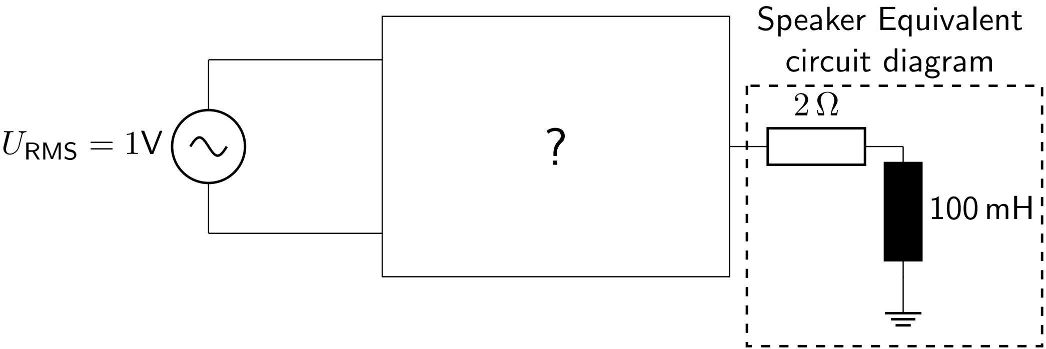
Negative feedback principle solution to the example problem (1)

Solution

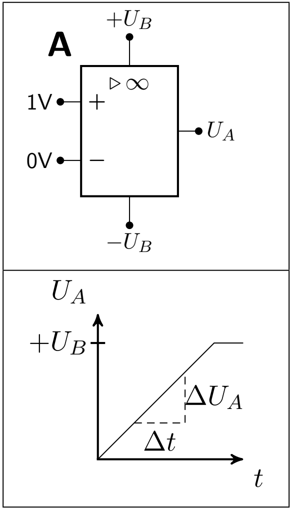
First, the voltage that should be present at the amplifier output must be calculated.

$$U_\mathsf{A} = \sqrt{P \cdot R} = \sqrt{25\ \mathrm{W} \cdot 4\ \Omega} = 10\ \mathrm{V} \tag{1}$$

How can an operational amplifier be used to replace the part of the circuit marked with a „?“ in order to achieve the required amplification of 5?



Negative feedback principle – solution to the example problem (2)



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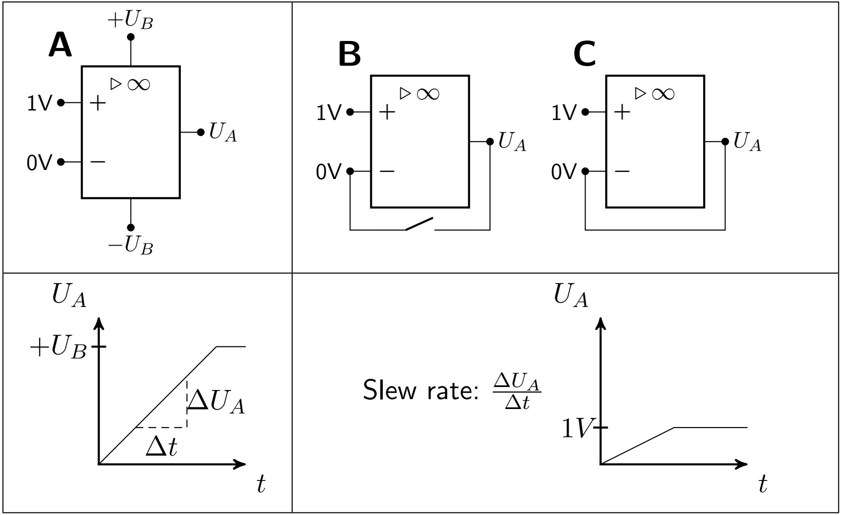

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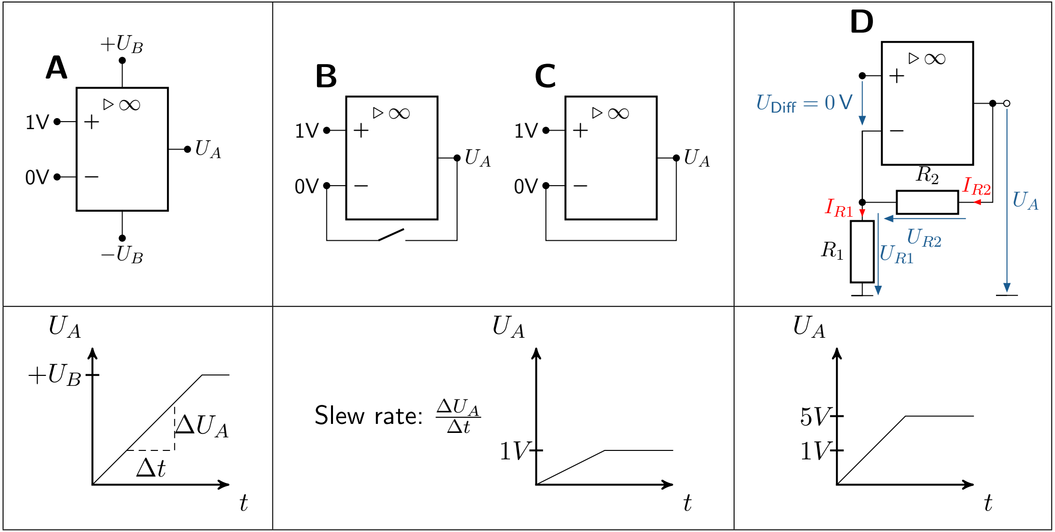
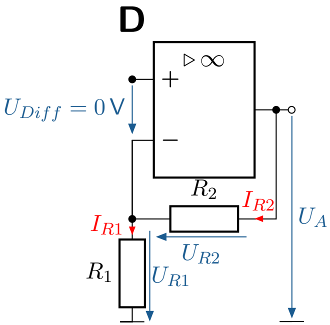


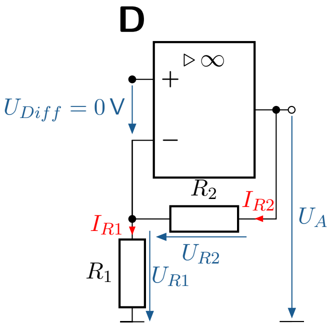
Principle of negative feedback - Solution to the example problem (3)

Assumption : $U_{\text{Diff}} = 0$

$$U_A = U_{R1} + U_{R2} = U_E + I_{R2} \cdot R_2 \qquad (2)$$



Final amplifier circuit with feedback



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Assumption : $U_{\text{Diff}} = 0$

$$U_A = U_{R1} + U_{R2} = U_E + I_{R2} \cdot R_2 \tag{2}$$

$$U_A = U_E + \frac{U_E}{R_1} \cdot R_2 \tag{3}$$

Principle of negative feedback - Solution to the example problem (3)

Assumption : $U_{\text{Diff}} = 0$

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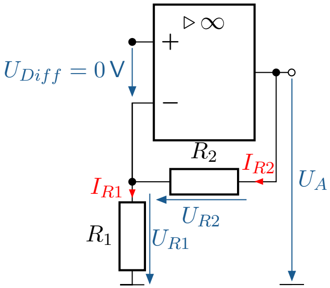


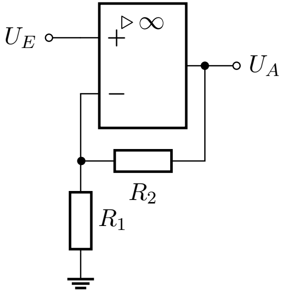
Final amplifier circuit with feedback

$$U_A = U_{R1} + U_{R2} = U_E + I_{R2} \cdot R_2 \qquad (2)$$

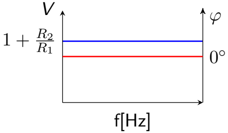
$$U_A = U_E + \frac{U_E}{R_1} \cdot R_2 \qquad (3)$$

$$\frac{U_A}{U_E} = \underbrace{(1 + \frac{R_2}{R_1})}_{V_{Gain}} \qquad (4)$$

Further OPV basic circuits can be found in the script.



(a) Circuit of a non-inverting amplifier



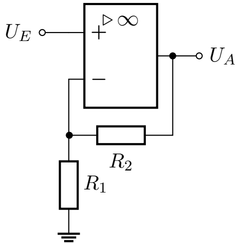
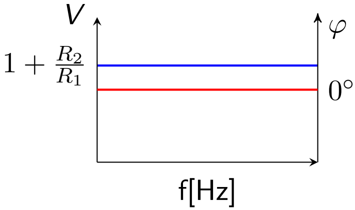
(b) Frequency response of a non-inverting amplifier

- Equation:

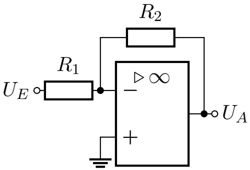
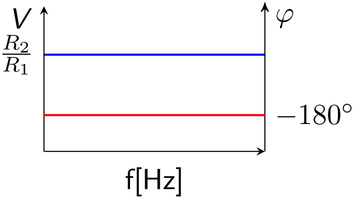
$$U_\mathsf{A} = \left(1 + \frac{R_2}{R_1}\right) U_\mathsf{E}$$

- Phase-aligned output signal
- Very high input resistance
- Low output resistance
- Applications: e.g., impedance converter

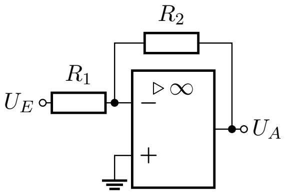
Non-inverting amplifier

Circuit	Frequency response and equation	Explanation and properties
	 $U_\mathrm{A} = \left(1 + \frac{R_2}{R_1}\right) U_\mathrm{E}$	- Phase-aligned output signal - Very high input resistance - Low output resistance - Applications: e.g., impedance converters

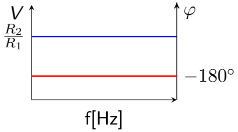
Non-inverting amplifier

Circuit	Frequency response and equation	Explanation and properties
	 $U_{\mathsf{A}} = \frac{R_2}{R_1} U_{\mathsf{E}}$	▶ -180° phase shift between input and output ▶ R_1 determines the input resistance ▶ Low-impedance output resistance ▶ Applications: e.g., active voltage divider for measuring high voltages when $R_1 > R_2$

New inverting amplifier



(a) Circuit of an inverting amplifier



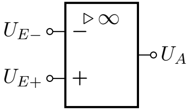
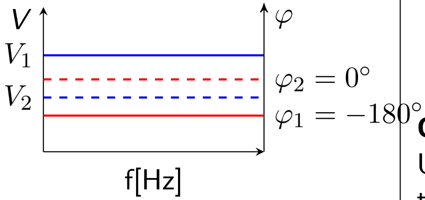
(b) Frequency response of an inverting amplifier

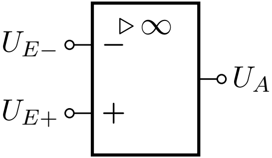
- Equation:

$$U_{\mathsf{A}} = \frac{R_2}{R_1} U_{\mathsf{E}}$$

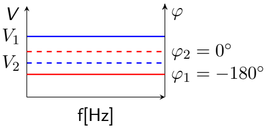
- -180° phase shift between input and output
- R_1 determines the input resistance
- Low-impedance output resistance
- Applications: e.g., active voltage divider for measuring high voltages when $R_1 > R_2$

Comparator

Circuit	Frequency response and equation	Explanation and properties
	 $V_1 : U_{\mathrm{E+}} > U_{\mathrm{E-}} \Rightarrow U_{\mathrm{A}} \approx +U_{\mathrm{B}}$ $V_2 : U_{\mathrm{E+}} < U_{\mathrm{E-}} \Rightarrow U_{\mathrm{A}} \approx -U_{\mathrm{B}}$	**Comparator** Use in two-point controllers and analog-to-digital converters



(a) Circuit of a comparator



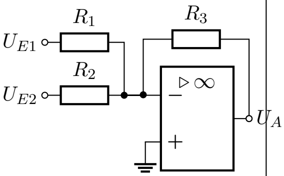
(b) Frequency response of a comparator

- Equation:

$$V_1 : U_{\text{E+}} > U_{\text{E-}} \Rightarrow U_{\text{A}} \approx +U_{\text{B}}$$

$$V_2 : U_{\text{E+}} < U_{\text{E-}} \Rightarrow U_{\text{A}} \approx -U_{\text{B}}$$

- Use in two-point controllers and analog-to-digital converters

<table>
<tr><th>Circuit</th><th>Frequency response and equation</th><th>Explanation and properties</th></tr>
<tr>
<td>
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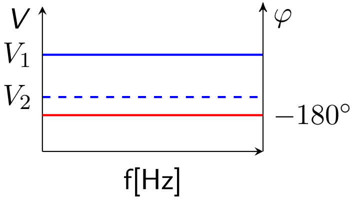
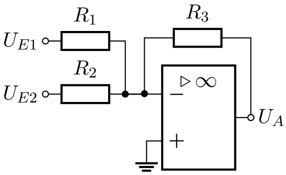
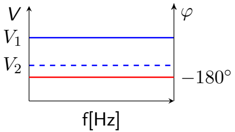
<td>

$$U_\mathrm{A} = \underbrace{\frac{R_3}{R_1}}_{V_1} \cdot U_\mathrm{E1} + \underbrace{\frac{R_3}{R_2}}_{V_2} \cdot U_\mathrm{E2}$$</td>
<td>The summing amplifier is based on the inverting amplifier and is used in analog computers and for mixing voltage signals.</td>
</tr>
</table>



(a) Circuit of a summing device

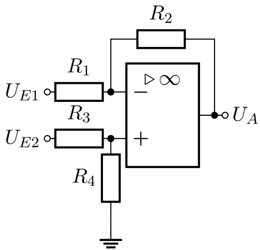
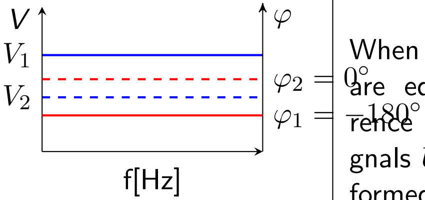


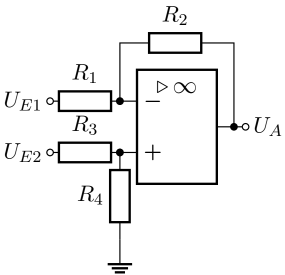
(b) Frequency response of a summing amplifier

- Gleichung:

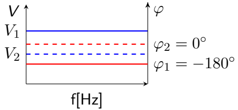
$$U_{\text{A}} = \underbrace{\frac{R_3}{R_1}}_{V_1} \cdot U_{\text{E1}} + \underbrace{\frac{R_3}{R_2}}_{V_2} \cdot U_{\text{E2}}$$

- The summing amplifier is based on the inverting amplifier and is used in analog computers and for mixing voltage signals.

Circuit	Frequency response and equation	Explanation and properties
	 $U_A = U_{\mathrm{E2}} \cdot \underbrace{\frac{R_1 + R_2}{R_1} \cdot \frac{R_4}{R_3 + R_4}}_{V_2} - U_{\mathrm{E1}} \cdot \underbrace{\frac{R_2}{R_1}}_{V_1}$	When all resistances are equal, the difference between the signals U_{E1} and U_{E2} is formed, which is why this circuit is referred to as a differential amplifier.



(a) Circuit of a subtractor

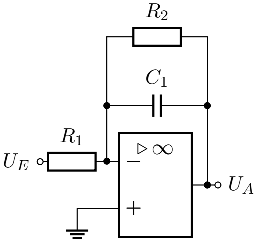
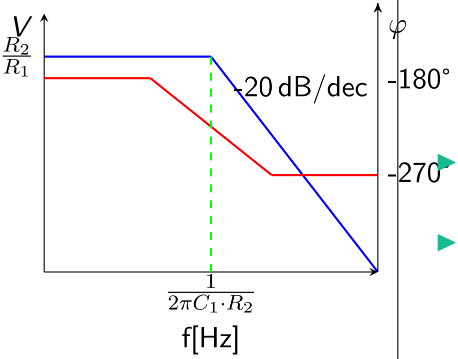


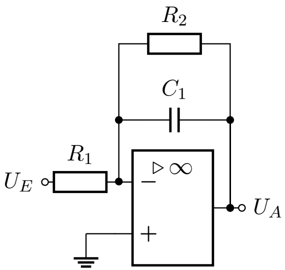
(b) Frequency response of a subtractor

- Equation:

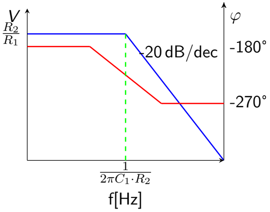
$$U_A = U_{\mathrm{E2}} \cdot \underbrace{\frac{R_1 + R_2}{R_1} \cdot \frac{R_4}{R_3 + R_4}}_{V_2} - U_{\mathrm{E1}} \cdot \underbrace{\frac{R_2}{R_1}}_{V_1}$$

- When all resistances are equal, the difference between the signals U_{E1} and U_{E2} is formed, which is why this circuit is referred to as a differential amplifier.

Circuit	Frequency response and equation	Explanation and properties
	 $U_A(t) = -\frac{R_2}{R_1}U_{\mathrm{E}}(t) - \int\limits_0^t \frac{U_{\mathrm{E}}(t)}{R_1C_1}\,dt$	- Integrates the input signal - Used as an active low-pass filter



(a) Circuit of an integrator

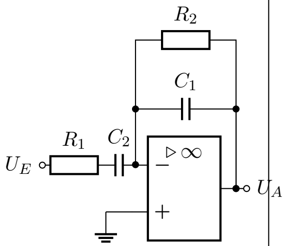
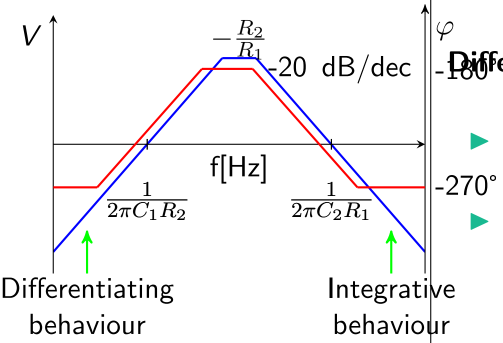


(b) Frequency response of an integrator

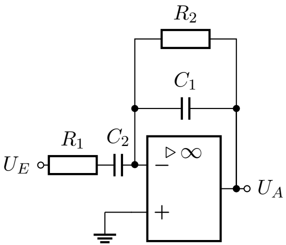
- Equation:

$$U_A(t) = -\frac{R_2}{R_1} U_{\mathsf{E}}(t) - \int_0^t \frac{U_{\mathsf{E}}(t)}{R_1 C_1}\, dt$$

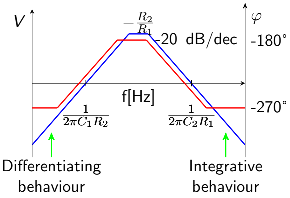
- Integrates the input signal
- Used as an active low-pass filter

Circuit	Frequency response and equation	Explanation and properties
	 $U_A = -\frac{R_2}{R_1} U_{\mathsf{E}}(t)$ $-\int^{t} \frac{1}{\ } U_{\mathsf{E}}(t)\,dt - R_2 C_2 \frac{dU_{\mathsf{E}}(t)}{\ }$	**Differentiator** - Integrates the input signal - Used as an active high-pass filter (in reality, it usually exhibits band-pass behavior, as shown here))

Differentiator new



(a) Circuit of a differentiator



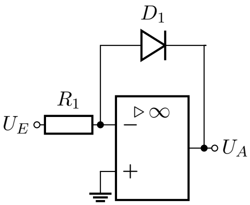
(b) Frequency response of a differentiator

- Equation:

$$U_\mathsf{A}(t) = -\frac{R_2}{R_1}\,U_\mathsf{E}(t) - \int_0^t \frac{1}{R_1 C_1}\,U_\mathsf{E}(\tau)\,d\tau -$$

- Integrates the input signal
- Used as an active high-pass filter (in reality, usually exhibits band-pass behavior, as shown here)

Logarithm calculator

Circuit	Frequency response and equation	Explanation and properties
	$U_{\mathsf{A}} = -U_{\mathsf{T}} \cdot \ln\left(\frac{U_{\mathsf{E}}}{R_1 \cdot I_S}\right)$ $U_{\mathsf{T}} = \frac{k_{\mathsf{B}} \cdot T}{e}$ $e =$ Elementary charge $k_{\mathsf{B}} =$ Boltzmann constant $I_{\mathsf{s}} =$ Reverse voltage of the diode	Forms the natural logarithm of the input signal

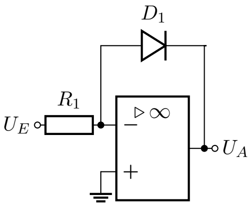


Figure: Circuit of a logarithmic converter

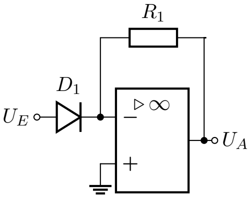
- Equation:

$$U_\mathsf{A} = -U_\mathsf{T} \cdot \ln\left(\frac{U_\mathsf{E}}{R_1 \cdot I_S}\right)$$

$$U_\mathsf{T} = \frac{k_\mathsf{B} \cdot T}{e}$$

$e =$ Elementary charge
$k_\mathsf{B} =$ Boltzmann constant
$I_\mathsf{s} =$ Reverse voltage of the diode

- Forms the natural logarithm of the input signal

Circuit	Frequency response and equation	Explanation and properties
	$U_\mathsf{A} = -R_1 \cdot I_\mathsf{S} \cdot e^{\frac{U_\mathsf{E}}{U_\mathsf{T}}}$	Has an e-functional relationship between input and output voltage

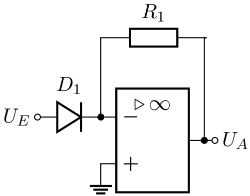


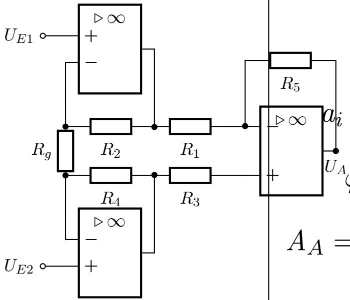
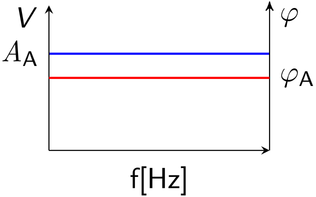
Figure: Circuit of a potentiometer

- Equation:

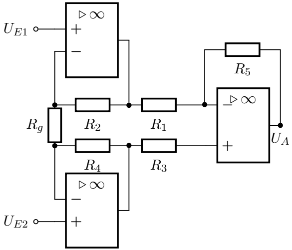
$$U_\mathsf{A} = -R_1 \cdot I_\mathsf{S} \cdot e^{\frac{U_\mathsf{E}}{U_\mathsf{T}}}$$

- Has an e-functional relationship between input and output voltage

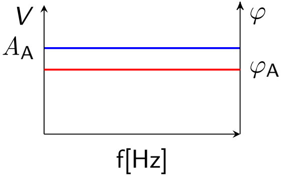
Instrument amplifier

Circuit	Frequency response and equation	Explanation and properties
	 a_i : Amplitude input voltage i φ_i : Phase input voltage i $A_A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos(\varphi_1 - \varphi_2)}$ $\tan(\varphi_A) = \frac{a_1 \sin(\varphi_1) + a_2 \sin(\varphi_2)}{}$	Differential amplifier with high input impedance and high common-mode rejection

New instrument amplifier



(a) Circuit of an instrument amplifier



(b) Frequency response of an instrument amplifier

- Equation:

$$a_i : \text{Amplitude input voltage i}$$

$$\varphi_i : \text{Phase input voltage i}$$

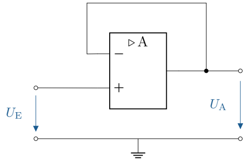
$$A_A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos(\varphi_1 - \varphi_2)}$$

$$\tan(\varphi_A) = \frac{a_1\sin(\varphi_1) + a_2\sin(\varphi_2)}{a_1\cos(\varphi_1) + a_2\cos(\varphi_2)}$$

$$\text{When} : R_2 = R_4 = R$$

$$U_\text{A} = (1 + \frac{2R}{R_\text{g}}) \cdot \frac{R_3}{R_2}(U_\text{E2} - U_\text{E1})$$

- Differential amplifier with high input impedance and high common-mode rejection

Circuit	Frequency response and equation	Explanation and properties
	Output voltage U_{A} follows input voltage U_{E} Gain $V = \frac{U_{\mathrm{A}}}{U_{\mathrm{E}}} = 1$	Also known as voltage followers High input resistance due to operational amplifiers Typical application: sensorsn

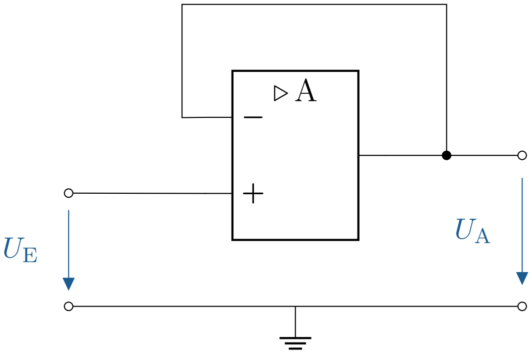


Figure: Circuit of an impedance converter

- Output voltage U_{A} follows the input voltage U_{E}
- Gain $V = \frac{U_{\mathrm{A}}}{U_{\mathrm{E}}} = 1$
- Also called a voltage follower
- High input resistance due to operational amplifier
- Typical application: Sensors

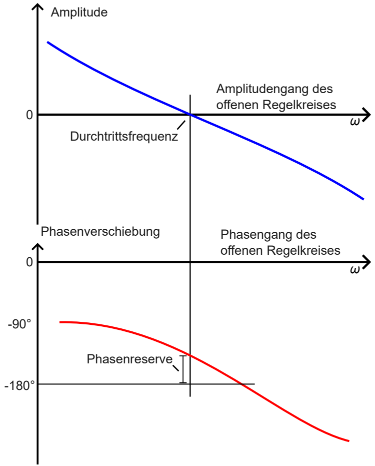
Stability of the operational amplifier control loop

The feedback of the output signal results in a control loop. This must be checked for stability. Three methods are presented for this purpose:

- Frequency response analysis based on the Bode diagram
- Examination of the transfer function for its pole locations
- Analysis of the transient response

Frequency response analysis (experimental/simulative)

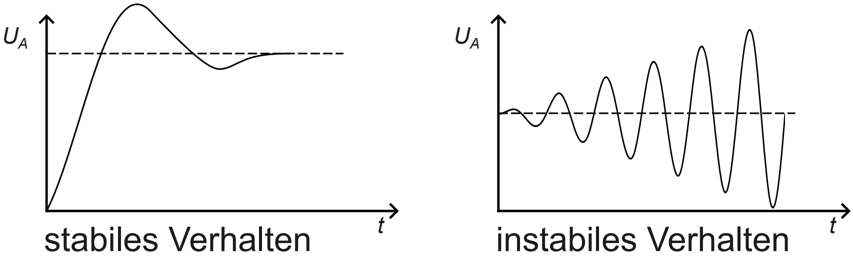
With this method, the Bode diagram is examined in the relevant frequency range.



On this basis, the phase position at the 0 dB pass point of the amplitude is determined. If there is a small distance to -180°, the system is unstable.

Analysis of the transfer function (analytical)

In this method, the circuit is stimulated with an input signal and the output behaviour is observed.



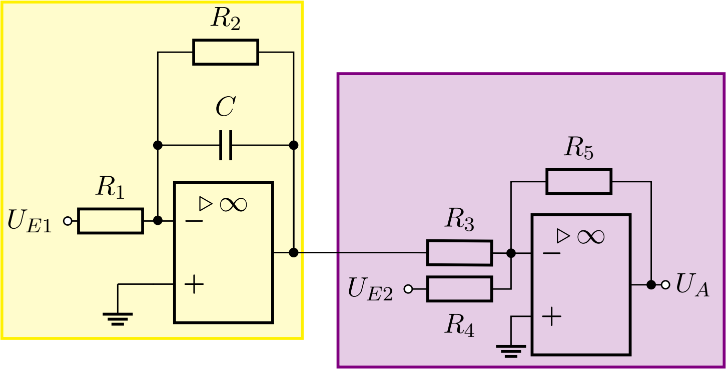
A circuit is to be designed that implements the following function:

$$U_{\mathsf{A}} = \int U_{\mathsf{E1}} + x \cdot U_{\mathsf{E1}} - 2 \cdot U_{\mathsf{E2}} \tag{5}$$

How can this be achieved? First, consider which basic circuits need to be combined to achieve this.

Solution step 1: Amplifier circuit

First, the above equation must be broken down into two sub-problems that can be solved using operational amplifier circuits. The first sub-problem is the integration of the input signal U_{E1}. To do this, an integrator circuit should first be used. The second sub-problem is the addition of the signals. A summing amplifier can be used for this.



The desired behaviour can therefore be achieved by combining an integrator circuit and a summing circuit. Is it possible to select x=0 with this circuit?

$$U_\mathsf{A} = \underbrace{\left[-\frac{1}{R_1 \cdot C} \int\limits_0^t U_\mathsf{E1}\; dt - \frac{R_2}{R_1} \cdot U_\mathsf{E1}\right]}_{Formel\ des\ Integrators} \underbrace{\cdot \left(-\frac{R_5}{R_3}\right) - \frac{R_5}{R_4} \cdot U_\mathsf{E2}}_{Formel\ des\ Summierers} \tag{6}$$

This can now be transformed as follows

$$U_\mathsf{A} = \underbrace{\frac{R_5}{R_1 \cdot R_3 \cdot C}}_{\overset{!}{=}1} \int\limits_0^t U_\mathsf{E1}\; dt + \underbrace{\frac{R_2 R_5}{R_1 R_3}}_{\overset{!}{=}x} U_\mathsf{E1} - \underbrace{\frac{R_5}{R_4}}_{\overset{!}{=}2} \cdot U_\mathsf{E2} \tag{7}$$

As can be seen from the formula, the component values now only need to be selected so that the correct pre-factors are obtained. A choice of x=0 is only possible if the resistor R_2 is omitted.